$F_X(x) = P(X \le x)$

Find $P(X \le s) = F_X(s)$

$\frac{d}{dx} F_X(x) = f_X(x)$

$\int_{-\infty}^{s} f_X(x)$

STAT 131: Lecture Notes 7

Winter 2024

7.1 Expectation [DS 4.1-4.2]

The distribution of a random variable tells us everything we may want to know about the behaviour of the random variable. However, we may want a more succinct summary of the random variable. Similar to how you might compute an average of a list of numbers to summarize them, we will discuss the average of a distribution, which is known as an expected value.

Example 7.1. Suppose you are an investor choosing between two startups to invest in. Startup A has a 10% chance of earning you $10,000,000 and a 90% chance of losing you $1,000,000. Meanwhile, Startup B has 50% chance of earning you $1,000,000 and a 50% chance of losing you $500,000. How would you choose which startup to invest in?

Startup A: $10,000,000 \times 0.1 + (-1,000,000) \times 0.9 = 100,000$

Startup B: $1,000,000 \times 0.5 + (-500,000) \times 0.5 = 250,000$

♢

Definition 7.1 (Expectation of a discrete random variable)**.** The expectation of a discrete random variable X, is the sum of the possible values of X, each weighted by its probability:

$$\mathbb{E}[X] = \sum_x x\mathbb{P}(X = x) = \sum_x x f(x)$$

as long as the sum is finite. The mean of the distribution is another term for $\mathbb{E}[X]$.

Example 7.2. Suppose $X \sim \text{Bernoulli}(p)$. What is $\mathbb{E}[X]$?

$$E[X] = \sum_x x \cdot P(X = x) = 0 \cdot (1-p) + 1 \cdot p = p$$

♢

Example 7.3. Suppose a disk has probability of 1/3 of failing each year. Let X be the random variable that represents the number of years until the disk fails, with pmf $f(x) = \frac{1}{3}\left(1 - \frac{1}{3}\right)^{x-1}$. What is the average number of years that it takes for the disk to fail?

$$E[X] = \sum_x x \cdot P(X=x) = \sum_{x=1}^{\infty} x \cdot \frac{1}{3}(1-1/3)^{x-1} = 1/3 \sum_{x=1}^{\infty} x\,(2/3)^{x-1}$$

$$= 1/3\left(1 + 2\cdot\frac{2}{3} + 3\left(\frac{2}{3}\right)^2 + 4\left(\frac{2}{3}\right)^3 + \cdots\right)$$

$$= 1/3\left(\frac{1}{(1-2/3)^2}\right) = 1/3\left(\frac{1}{1/3\cdot 1/3}\right) = 1/(1/3) = \boxed{3}$$

♢

Example 7.4. Suppose $X \sim \text{Poisson}(\lambda)$. What is $\mathbb{E}[X]$?

$$E[X] = \sum_x x \cdot P(X=x) = \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda}\lambda^x}{x!} = \sum_{x=1}^{\infty} x\,\frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda}\sum_{x=1}^{\infty} \lambda^x \frac{x}{x!} = e^{-\lambda}\sum_{x=1}^{\infty} \lambda^x \frac{1}{(x-1)!} = \lambda e^{-\lambda}\sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$= \lambda e^{-\lambda}\sum_{x=0}^{\infty} \frac{\lambda^x}{x!} = \lambda e^{-\lambda} e^{\lambda} = \boxed{\lambda}$$

♢

In some cases, the expectation of a random variable may not exist (i.e., the sum may not converge).

Example 7.5. Suppose X is a random variable with pmf $f(x) = \frac{6}{\pi^2 x^2}$ for $x = 1, 2, 3, \ldots$. Then $\mathbb{E}[X]$ does not exist.

$$\sum_x x \cdot P(X=x) = \sum_{x=1}^{\infty} x \frac{6}{\pi^2 x^2} = \frac{6}{\pi^2}\sum_{x=1}^{\infty} \frac{x}{x^2} = \frac{6}{\pi^2}\sum_{x=1}^{\infty} \frac{1}{x}$$

but $\lim_{n\to\infty} \sum_{x=1}^{n} \frac{1}{x} = \infty$

♢

Definition 7.2 (Expectation of a continuous random variable)**.** The expectation of a continuous random variable X, is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x)dx$$

as long as the integral is finite.



Example 7.6. Suppose $X \sim \text{Uniform}(a, b)$. What is $\mathbb{E}[X]$?

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\,dx = \int_a^b x \cdot \frac{1}{b-a}\,dx = \frac{x^2}{2(b-a)}\Big|_a^b$$

$$= \frac{b^2 - a^2}{2(b-a)} = \frac{(b-a)(b+a)}{2(b-a)} = \boxed{\frac{b+a}{2}}$$

♢

Example 7.7. Suppose $X \sim \text{Exponential}(\lambda)$. What is $\mathbb{E}[X]$?

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x)\,dx = \int_0^{\infty} x\,\lambda e^{-\lambda x}\,dx$$

$u = \lambda x \quad dv = e^{-\lambda x}\,dx$

$du = \lambda\,dx \quad v = -\frac{1}{\lambda} e^{-\lambda x}$

$$= vu\Big|_0^{\infty} - \int_0^{\infty} v\,du = -x\,e^{-\lambda x}\Big|_0^{\infty} - \int_0^{\infty} -e^{-\lambda x}\,dx$$

$$= (0-0) - \left(\frac{1}{\lambda} e^{-\lambda x}\Big|_0^{\infty}\right) = -\left(0 - \frac{1}{\lambda}\right) = \frac{1}{\lambda}$$

♢

The expectation of the transformation of a random variable can be computed by the Law of the Unconscious Statistician (LOTUS).

Theorem 7.1 (LOTUS)**.** *Suppose X is discrete. Then for any function $h(\cdot)$,*

$$\mathbb{E}[h(X)] = \sum_x h(x) f(x),$$

if the sum is finite.
Suppose X is continuous. Then for any function $h(\cdot)$,

$$\mathbb{E}[h(X)] = \int_{-\infty}^{\infty} h(x) f(x)\,dx,$$

if the integral is finite.

Example 7.8 (Volume of a sphere)**.** Consider a sphere, where the radius is a random variable R, with

$$f(r) = \mathbb{P}(R = r) = \frac{1}{3} \text{ for } r = 1, 2, 3.$$

What is the expected value of the volume of the sphere?

$$E[\text{volume}] = E\left[\tfrac{4}{3}\pi R^3\right] = \sum_r \tfrac{4}{3}\pi r^3 \cdot P(R=r)$$

$$= \tfrac{4}{3}\pi 1^3 \cdot P(R=1) + \tfrac{4}{3}\pi 2^3 \cdot P(R=2) + \tfrac{4}{3}\pi 3^3 \cdot P(R=3)$$

$$= \tfrac{1}{3}\left(\tfrac{4}{3}\pi 1^3 + \tfrac{4}{3}\pi 2^3 + \tfrac{4}{3}\pi 3^3\right) = 16\pi$$

♢

Example 7.9. Suppose X is a random variable with $\mathbb{P}(X = 0) = 0.2$, $\mathbb{P}(X = 1) = 0.5$, and $\mathbb{P}(X = 2) = 0.3$. What is $\mathbb{E}[X]$? What is $\mathbb{E}[2X^2 + 3]$?

$$E[X] = \sum_x x \cdot P(X=x) = 0 \cdot (0.2) + 1 \cdot (0.5) + 2 \cdot (0.3) = 1.1$$

$$E[2X^2+3] = \sum_x (2x^2+3) \cdot P(X=x) = (2 \cdot 0^2+3) \cdot (0.2) + (2 \cdot 1^2+3) \cdot (0.5) + (2 \cdot 2^2+3) \cdot (0.3)$$

$$= 6.4$$

♢

Example 7.10. Suppose $X \sim \text{Uniform}(a, b)$. What is $\mathbb{E}[X^2]$?

$$E[X^2] = \int_{-\infty}^{\infty} x^2 \cdot f(x)\, dx = \int_a^b \frac{x^2}{b-a}\, dx = \frac{x^3}{3(b-a)}\Big|_a^b$$

$$= \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(b^2+ab+a^2)}{3(b-a)}$$

♢

$$= \frac{b^2+ab+a^2}{3}$$

Example 7.11. Suppose the failure rate X for a server has pdf

$$f(x) = 3x^2. \text{ for } 0 < x < 1.$$

What is the expected value of $Y = 1/X$, the time until failure?

$F_X(x) = \int_0^x 3t^2\,dt = x^3$

$$E[Y] = E[1/X] = \int_{-\infty}^{\infty} \frac{1}{x} f(x)\,dx = \int_0^1 \frac{1}{x} 3x^2\,dx = \int_0^1 3x\,dx$$

$$= \frac{3}{2}x^2\Big|_0^1 = 3/2$$

Hard way $P(Y \le y) = P(1/X \le y) = P(X \ge 1/y) = 1 - P(X \le 1/y) = 1 - F_X(1/y) = 1 - (1/y)^3 = F_Y(y)$

$f_Y(y) = \frac{d}{dy}\,1 - y^{-3} = 3y^{-4}$ for $y > 1$

$$\Rightarrow E[Y] = \int_{-\infty}^{\infty} y f_Y(y)\,dy = \int_1^{\infty} y\,3y^{-4}\,dy = 3/2$$

♢

The LOTUS can also be extended to functions of two different random variables X and Y.

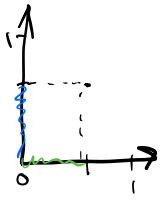
Theorem 7.2 (Bivariate LOTUS)**.** *Suppose X and Y are discrete. Then for any function $h(\cdot)$,*

$$\mathbb{E}[h(X,Y)] = \sum_x \sum_y h(x,y) f(x,y),$$

if the sum is finite.
Suppose X and Y are continuous. Then for any function $h(\cdot)$,

$$\mathbb{E}[h(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x,y) f(x,y)\,dx\,dy,$$

if the integral is finite.



Example 7.12. Suppose a point (X,Y) follows the uniform distribution on the square $[0,1]^2$. Notice that X and Y make up two sides of a rectangle, and their product XY is the area of the rectangle. Find $\mathbb{E}[XY]$, the expected area of the rectangle.

$f(x,y) = 1$ for $0 < x < 1$ and $0 < y < 1$ $\quad h(x,y) = xy$

$$E[XY] = \int_0^1 \int_0^1 h(x,y) \cdot f(x,y)\,dx\,dy = \int_0^1 \int_0^1 xy\,dx\,dy = \int_0^1 \left(\frac{x^2 y}{2}\Big|_0^1\right) dy = \int_0^1 \frac{y}{2}\,dy$$

$$= \frac{y^2}{4}\Big|_0^1 = \boxed{1/4}$$

♢

7.1.1 Properties of Expectation

Theorem 7.3 (Linearity). *Expectation of a linear function of a single random varaible X:*

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

Expectation of a linear function of many random variables $X_1, X_2, \ldots, X_n$:

$$\mathbb{E}\left[\sum_{i=1}^{n} a_i X_i + b_i\right] = \sum_{i=1}^{n} a_i \mathbb{E}[X_i] + b_i$$

Example 7.13. Let $X \sim \text{Binomial}(n, p)$. What is $\mathbb{E}[X]$?

X = # of successes in n trials $= X_1 + X_2 + \cdots + X_n$ where $X_i \sim \text{Bernoulli}(p)$

$$E[X] = E[X_1] + E[X_2] + \cdots + E[X_n]$$
$$= p + p + \cdots + p = \boxed{np}$$

♢

Example 7.14. At a party, n people put their drink on a table. Later in the evening, nobody remembers which cup is theirs, so they just each grab a cup at random. Let X denote the number of people who pick back up their own cup. What is $\mathbb{E}[X]$? Do you think $\mathbb{E}[X]$ will be larger when n is big or small?

$X = Z_1 + Z_2 + \cdots + Z_n$ where $Z_i = \begin{cases} 1, & \text{if the } i^{\text{th}} \text{ person gets their cup} \\ 0, & \text{otherwise} \end{cases}$

$$E[X] = E[Z_1] + E[Z_2] + \cdots + E[Z_n]$$
$$= n\,E[Z_i] \qquad \text{but } E[Z_i] = \left(1 \cdot \frac{1}{n} + 0 \cdot \frac{(n-1)}{n}\right) = \frac{1}{n}$$
$$= n \cdot \frac{1}{n} = \boxed{1}$$

Doesn't depend on n!

♢

Theorem 7.4. *Let $X_1, \ldots X_n$ be independent random variables. Then*

$$\mathbb{E}[\prod_{i=1}^{n} X_i] = \prod_{i=1}^{n} \mathbb{E}[X_i].$$

A special case of this tells us that for random variables X and Y, where $X \perp Y$,

$$\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y].$$

Example 7.15 (Question?)**.** If $\mathbb{E}[XY] = \mathbb{E}[X] \cdot \mathbb{E}[Y]$ does that mean that $X \perp Y$?

No

Consider the following example.

$P(X=1) = 1/3$, $P(X=0) = 1/3$, $P(X=-1) = 1/3$

and let $Y = X^2$

Then $P(Y=1) = 2/3$ and $P(Y=0) = 1/3$ ◊

Notice that $P(X=1, Y=1) = 1/3 \neq P(X=1) \cdot P(Y=1)$

So X and Y are not indep.! ☆

but $E[XY] = E[X X^2] = E[X^3] = 0$

and $E[X] = 0 \Rightarrow E[X] \cdot E[Y] = 0$

So $E[XY] = E[X] \cdot E[Y]$ ☆